

check before final draft.

Mathematics Exploration ☒ Student Checklist

1. Is your exploration written entirely by yourself – during which you avoided simply replicating work and ideas from sources you found during your research? ☐ Yes ☐ No
2. Have you strived to apply your personal interest; develop your own ideas; and use critical thinking skills when writing your exploration? ☐ Yes ☐ No
3. Have you endeavoured to address all five assessment criteria? ☐ Yes ☐ No
4. Does your exploration focus on good mathematical communication – and read like an article for a mathematical journal? ☐ Yes ☐ No
5. Have you included an introduction and conclusion, and clearly expressed your aim(s)? ☐ Yes ☐ No
6. Have you documented all of your source material in a detailed bibliography in line with the IB academic honesty policy? ☐ Yes ☐ No
7. Not including the bibliography, is your exploration in the range of 12 to 20 pages? ☐ Yes ☐ No
8. Are graphs, tables and diagrams sufficiently described and labelled? ☐ Yes ☐ No
9. Have you used and demonstrated mathematics that is at the same, or similar, level of the mathematics in your IB mathematics course? ☐ Yes ☐ No
10. Have you attempted to discuss mathematical ideas, and use mathematics, with a sufficient level of knowledge and understanding demonstrated on your part? ☐ Yes ☐ No
11. Are formulae, graphs, tables and diagrams in the main body of text? (preferably no full-page graphs; and no separate appendices) ☐ Yes ☐ No
12. Have you used technology – such as a GDC, spreadsheet, mathematics software, drawing or word-processing software – to enhance mathematical communication? ☐ Yes ☐ No
13. Have you used appropriate mathematical language (notation, symbols, terminology) and defined key terms? ☐ Yes ☐ No
14. Is the mathematics in your exploration performed precisely and accurately? ☐ Yes ☐ No
15. Has calculator/computer notation and terminology **not** been used?
($y = x^2$, not $y = x^{\wedge}2$; $\approx$, not $=$ for approx. values; π , not pi; $|x|$, not abs(x); etc) ☐ Yes ☐ No
16. Have you included – throughout your exploration – reflective and explanatory comments about the mathematics being explored and any results obtained? ☐ Yes ☐ No

Criteria for the Exploration (IA)

Criterion A: Presentation (4 marks)

This criterion assesses the organization and coherence of the exploration. A well-organised exploration has an **introduction**, a **rationale** (a brief explanation of why the topic was chosen), describes the **aim** of the exploration and has a **conclusion**. A coherent exploration is logically developed and easy to follow.

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	The exploration has some coherence or some organization.
2	The exploration has some coherence and shows some organization.
3	The exploration is coherent and well organized.
4	The exploration is coherent, well organized, and concise.

Further Guidance

- A **complete** exploration will have all steps clearly explained, and will meet its aim.
- **Organization** refers to the overall structure or framework, including the introduction, body, conclusion etc.
- A **coherent** exploration displays a logical development and is not difficult to follow ('reads well').
- A **concise** exploration remains focused on the overall aim and avoids irrelevant material.
- Key ideas and concepts need to be clearly explained.
- Graphs, tables and diagrams should be embedded in the text where most appropriate and not be put in an appendix at the end of the document.
- The use of technology is not required but strongly encouraged where appropriate.
- It is absolutely critical that the use of a source is cited (footnoted) at the location where it is used.
- Your bibliography must list all sources (books, websites, etc) you consulted when writing your Exploration.

Criterion B: Mathematical Communication (4 marks)

This criterion assesses to what extent you are able to clearly and effectively use multiple forms of mathematical representation such as formulae, diagrams, tables, graphs and models.

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	The exploration contains some relevant mathematical communication which is partially appropriate.
2	The exploration contains some relevant appropriate mathematical communication.
3	The mathematical communication is relevant, appropriate and mostly consistent.
4	The mathematical communication is relevant, appropriate and consistent throughout.

Further Guidance

- You must use appropriate mathematical language (notation, symbols & terminology).
- Use multiple forms of mathematical representation – such as formulae, diagrams, tables, charts, graphs and models, where appropriate.
- The meaning of key terms should be clear and any variables or parameters should be explicitly defined.
- Use a deductive method and set out proofs logically where appropriate.
- All graphs, tables & diagrams should be clearly labelled – and include captions where appropriate.
- Do not use calculator or computer notation unless it is software generated and cannot be changed.

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Criterion C: Personal Engagement (3 marks)

This criterion assesses the extent you engage with the exploration and make it your own. This includes independent thinking, creativity, addressing personal interest and presenting math ideas in your own way.

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	There is evidence of some personal engagement.
2	There is evidence of significant personal engagement.
3	There is evidence of outstanding personal engagement.

Further Guidance

- Personal engagement can include: thinking independently or creatively, presenting mathematical ideas in your own way, exploring the topic from different perspectives, or making and testing predictions.
- If it is necessary to include mathematical work from a source such as a textbook in your Exploration then you should endeavour to insert your own comments and description of the work as much as possible.
- Ways to show personal engagement include: investigating your own questions & conjectures; making up your own examples; presenting ideas & results in your own words; creating your own models or functions.

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Criterion D: Reflection (3 marks)

This criterion assesses how well you review, analyse and evaluate your exploration. Although reflection may be seen in the conclusion, it should also exist throughout the exploration. Reflection may be demonstrated by considering limitations or extensions or by relating mathematical ideas to your own previous knowledge.

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	There is evidence of limited reflection.
2	There is evidence of meaningful reflection.
3	There is substantial evidence of critical reflection.

Further Guidance

- Simply describing results represents **limited or superficial reflection**. To achieve a score higher than 1 you will need to provide deeper and more sophisticated consideration of methods and results.
- Ways of showing **meaningful reflection** include: linking results to the aim of your Exploration; commenting on what you have learned; considering limitations; or comparing different mathematical approaches.
- Ways of showing **critical reflection** include: considering implications of results; discussing strengths and weaknesses of methods; considering different perspectives; making links between different areas of math.
- Substantial evidence** means that critical reflection is present throughout the exploration.

Criterion E: Use of Mathematics (6 marks) * descriptors for SL and HL are not the same *

This criterion assesses to what extent you use mathematics that is relevant to your Exploration. The mathematics explored should either be part of the syllabus, at a similar level, or slightly beyond. You need to demonstrate that you understand the mathematics by providing clear comments and explanations, and illustrating with examples or applications. The mathematics can be regarded as correct even if there are a few minor errors so long as they do not detract from the flow of mathematics or lead to an incorrect or inaccurate result. Lines of reasoning must be shown to justify steps in the mathematical development of the exploration.

Criterion E: Use of Mathematics – SL

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	Some relevant mathematics is used.
2	Some relevant mathematics is used. Limited understanding is demonstrated.
3	Relevant mathematics commensurate with the level of the course is used. Limited understanding is demonstrated.
4	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is partially correct. Some knowledge and understanding are demonstrated.
5	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is mostly correct. Good knowledge and understanding are demonstrated.
6	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Thorough knowledge and understanding are demonstrated.

Further Guidance SL

- You must clearly demonstrate your understanding of the mathematics you write about in your Exploration. For knowledge and understanding to be **thorough** it must be demonstrated throughout.
- The mathematics only needs to be what is required to support the development of the exploration. It is better to do a few things well than a lot of things not so well. If the mathematics used is **relevant** to the topic being explored, **commensurate** with the level of the course and understanding is clearly demonstrated, then it can achieve a high level in this criterion.

Criterion E: Use of Mathematics – HL

Achievement Level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	Some relevant mathematics is used. Limited understanding is demonstrated.
2	Some relevant mathematics is used. The mathematics explored is partially correct. Some knowledge and understanding are demonstrated.
3	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Some knowledge and understanding are demonstrated.
4	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Good knowledge and understanding are demonstrated.
5	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct and demonstrates sophistication or rigour. Thorough knowledge and understanding are demonstrated.
6	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is precise and demonstrates sophistication and rigour. Thorough knowledge and understanding are demonstrated.

Further Guidance HL

- You must clearly demonstrate your understanding of the mathematics you write about in your Exploration. For knowledge and understanding to be **thorough** it must be demonstrated throughout.
- Sophistication in mathematics** may include understanding & use of challenging math concepts, looking at a problem from different perspectives and seeing underlying structures to link different areas of mathematics.
- Rigour** involves clarity of logic and language when making mathematical arguments and calculations.
- Precise mathematics** is error-free and uses an appropriate level of accuracy at all times.